Alyssa Lu Lee

PhD Student
Institute for Quantitative Biomedicine
Rutgers University – New Brunswick

alyssa.lee@rutgers.edu

EDUCATION

PhD **Quantitative Biomedicine**, Rutgers University, 2025–
Advised by Dr. Rohan Maddamsetti and Dr. Sagar Khare

BS **Computational Biology**, Carnegie Mellon University, 2020–2024

EXPERIENCE

2026–present **Research Fellow**
Institute for Quantitative Biomedicine
Rutgers University – New Brunswick

2024–2025 **Bioinformatics Analyst**
Bioinformatics Core, Department of Pediatrics, School of Medicine
University of Pittsburgh

2023 **Nonclinical Research and Development Intern**
Bristol Myers Squibb

2022–2023 **Undergraduate Research Assistant**
Machine Learning Department
Carnegie Mellon University

2022 **Software Engineer Intern**
CNH Industrial

RESEARCH AREAS

AI methods for protein design

Phage-host binding

PUBLICATIONS

Preprints and Submitted Manuscripts

2026 **Lee, A. L.**, Sharma, A., & Maddamsetti, R. (2026). *Zero-shot ecological annotation of microbial genomes with myLLannotator accelerates scientific discovery* (p. 2026.01.18.700140) [preprint]. *bioRxiv*. https://doi.org/10.64898/2026.01.18.700140

2026 Zaky, S. H., Rajasundaram, D., **Lee, A. L.**, Summersgill, K., Levine, S., Boulos, O., Intini, G., Sfeir, C., & Silva, R. M. (2026). *Using Spatial Transcriptomics to Identify a Diagnostic Molecular Signature Associated with Reversible Dental Pulpitis* [preprint]. Research Square. https://doi.org/10.21203/rs.3.rs-7955718/v1

2026 Jacoby, M. L., Sun, C., Farren, J. L., Marti, M. M., **Lee, A. L.**, Donovan, T., Klein, R., Rajasundaram, D., & Klimstra, W. B. (2026). *Biphasic lymphoid tissue activation drives systemic innate immune responses to alphavirus infection* [Manuscript submitted for publication].

Journal Articles

2026 Kargl, C. K., Nindl, B. C., Goulart, J. B., Feigel, E. D., Sahu, A., Fazeli, P. K., **Lee, A. L.**, Rajasundaram, D., Ambrosio, F., Hubal, M. J., Lovalekar, M. T., Martin, B. J., Sterczala, A. J., & Koltun, K. J. (2026). Varying Influence of Menstrual Cycle Phase and Hormonal Contraceptive Use on Resistance Exercise-Induced Circulating Extracellular Vesicle and Metabolite Signalling. *Journal of Extracellular Vesicles*, 15(3), e70244. https://doi.org/10.1002/jev2.70244

PRESENTATIONS

2026 **Lee, A. L.**, Sharma, A., & Maddamsetti, R. (2026, March 12). *myLLannotator: Using large language models for metadata annotation* [conference talk]. Rutgers Conference for Rising Stars in Graduate Research 2026, Rutgers University, New Brunswick, NJ.

2026 **Lee, A. L.**, Seamann, A. L., Chung, G., Zhao, C., Khare, S.D. (2026, April 7). *HaloMPNN: AI Redesign of Proteins for Salt Tolerance* [poster]. Sargassum Biorefinery Annual Meeting, Rutgers University, New Brunswick, NJ.

Updated May 2026